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PHA 101: Introduction to Pharmacy - Drugs, Dosage Forms and the Pharmacist's Role

Orientation to pharmacy practice, drug nomenclature and pharmaceutical calculations

FACULTY PRIMER

Faculty Primer

Faculty of Pharmaceutical Sciences
Pharmaceutics & Pharmaceutical Technology | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

This pack orients new pharmacy students to the profession, explains drug nomenclature and dosage forms, and drills the pharmaceutical calculations that catch students out in early assessments, including percentage concentrations, dilutions and dose calculations.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Explain the pharmacist's roles in dispensing, counselling, compounding and pharmacovigilance.
- Distinguish chemical, generic and brand names and discuss bioequivalence.
- Classify dosage forms and explain how the route of administration affects onset and bioavailability.
- Perform percentage, dilution and dose calculations accurately with correct units.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What pharmacy is for

Pharmacy is the science and practice of preparing and using medicines safely and effectively.

- Dispensing: verifying the prescription, preparing and labelling the medicine.
- Counselling: explaining dose, frequency, duration, storage and side effects to the patient.
- Compounding: extemporaneous preparation where no commercial product fits.
- Pharmacovigilance: detecting, assessing and reporting adverse drug reactions.
- Regulation: controlled drugs, prescription requirements and record keeping.
- Medication reconciliation and adherence support reduce avoidable harm.

The pharmacist's check

Right patient, right drug, right dose, right route, right time and right documentation. Any doubt is a reason to stop and verify, not to guess.

Names, dosage forms and bioavailability

Names and formulations are not trivia; they determine how a medicine behaves in the body.

- Chemical name describes structure; generic name is the internationally accepted non-proprietary name; brand name is the manufacturer's trade name.
- Bioavailability is the fraction of an administered dose that reaches systemic circulation unchanged.
- Bioequivalence means two products deliver the same rate and extent of absorption within accepted limits.
- Oral solid forms: tablets, capsules, granules. Modified-release designs alter release rate.
- Liquid forms: solutions, suspensions, emulsions; shake suspensions before measuring.
- Parenteral, topical, rectal, inhaled and ophthalmic routes each have distinct advantages and risks.

Calculations without panic

Set every calculation out in one consistent format: known, required, conversion, equation, answer with units.

- Percentage weight in volume and weight in weight are not interchangeable; read what the label states.
- Dilution: C1 x V1 = C2 x V2, with concentrations in the same unit.
- Dose by body weight: dose in mg per kg multiplied by weight in kg, then checked against the maximum adult dose.
- Dose by body surface area requires the stated formula and correctly converted height and weight.
- Alligation solves mixing problems between two strengths.
- Always sanity-check: an answer that implies 40 tablets per dose is almost certainly wrong.

Units decide safety

Milligrams and micrograms differ by a factor of 1000. A decimal slip can be fatal. Convert to the same unit before calculating and write the unit at every step.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. A prescription requires 250 mg and the stock is a 5 percent w/v solution. What volume is needed?

Answer: Five percent w/v is 5 g per 100 mL, or 50 mg per mL. 250 mg therefore requires 5 mL.

2. Dilute 100 mL of a 20 percent solution to 500 mL. What is the final concentration?

Answer: C1V1 = C2V2 gives 20 x 100 = C2 x 500, so C2 = 4 percent.

3. What is bioequivalence?

Answer: The absence of a significant difference in the rate and extent to which the active ingredient becomes available at the site of action between two products.

4. Why must a suspension be shaken before use?

Answer: The dispersed particles settle, so shaking redistributes them and ensures the measured dose is representative.

5. A child weighs 20 kg and the dose is 10 mg/kg per day in two divided doses. What is each dose?

Answer: The daily dose is 200 mg, divided into two doses of 100 mg each.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] List six roles of a pharmacist and one example of each.
- [ ] Explain chemical, generic and brand names in one sentence each.
- [ ] Make a table of dosage forms with two advantages and two disadvantages.
- [ ] Solve ten percentage and dilution calculations showing all units.
- [ ] Memorise the right-patient checking list.
- [ ] Practise weighing and measuring in the laboratory with a supervisor.
- [ ] Read the current NAFDAC or PCN guidance on dispensing and write five key points.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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